

This example illustrates what we have been trying to explain throughout this book: an analytical method must always undergo serious, documented, and careful validation before being used routinely, and even more so when it is used to compare measurements made in different laboratories.

In addition, this validation must allow the analyst to explain clearly to the decision-maker the risk of considering the result as a definitive element of verdict. The concept of measurement uncertainty allows this risk to be identified and, more importantly, quantified. But, for measurement uncertainty to be able to play an effective role in decision-making based on an analytical result, a certain number of questions need to be asked. But a quick review will show that many answers are not too difficult to get.

11.1 Role of the Number of Replicates

It was pointed out in Section 8.4.3 that replication can reduce MU and facilitate or enable decision-making when the coverage interval appears overly large for an unambiguous decision. Making replicates is an effective approach to reduce uncertainty and thus facilitate decision-making. But it is necessary to be rigorous in how replication is designed to avoid any computational bias in estimating MU.

The role of replicates, in terms of number and mode of obtention, remains an open question. This is due to the ambiguity of the definition of a replicate in terms of independence from another replicate, but not only. Depending on the starting point of the replication within the sequence of the analytical operating procedure, as chosen by the analyst, the estimation of the MU is different.

When replication is conducted in the same series or done under repeatability conditions, the calculation is relatively simple. But it can also be done in another series (e.g. another day), and therefore acquired under intermediate precision conditions. From a statistical point of view it is mandatory to consider the between-series effect, which is often poorly known or estimated.

Another way to reduce an overly-large MU is to calculate the uncertainty budget. This tool is briefly discussed in Section 6.6. It consists in quantifying the relative importance of each source of uncertainty. This is straight-forward when applied to a type B approach. However for a type A approach, it implies a significant experimental effort and a complex statistical treatment, also mentioned in Section 7.2.4. Once the uncertainty budget is available, the main source(s) of uncertainty can be identified, and ways to reduce them can be explored.

11.2 Traceability to International Units

In Section 6.3, it was underlined that the attachment of chemical measurements to the International System of Units (SI) is delicate. Indeed, the mole never had a figurative standard, as exists for the meter or the kilogram. There has never been a

simple method to ensure its traceability. To overcome this peculiarity of the analytical sciences, the Consultative Committee for the Quantity of Matter (CCQM) has proposed two solutions: primary methods and certified reference materials.

To achieve traceability to SI units, primary analytical methods were recommended; as an example, an approach based on isotopic dilution coupled with high-resolution mass spectrometry (HR-MS). But there are also titrimetry and gravimetry following very particular protocols, which are admitted to the exclusive group of primary methods. For example analyzing pesticides by liquid chromatography mass spectrometry (LC-MS) or liquid chromatography hyphenated to tandem mass spectrometry (LC-MS/MS) using labeled molecules can lead to a primary method.

Certified reference materials are the other less expensive alternative, as they do not always require very sophisticated instrumentation; they are also promoted by the CCQM. Various protocols for interlaboratory comparisons are proposed. The following table (Table 11.1) is an attempt to classify these studies. The production of CRMs is considered the most satisfactory from a metrological point of view.

It can be said that a CRM is a substitute for a primary standard comparable to what the kilogram or the meter were in the past. Given the huge diversity of measurements in the analytical sciences, this helps to understand why unambiguous traceability will remain difficult.

Table 11.1 Tentative classification of three interlaboratory comparison procedures.

Characteristics	Collaborative study	Interlaboratory analysis	Proficiency testing
Main goal	Estimate the most probable value	Estimate method precision	Estimate laboratory competence
Context	Certified material production	Method standard publication	Laboratory accreditation
Main references	BIPM Guide, ISO 17034, FD ISO Guide 35	ISO 5725 series	ISO 13528
Organizer	Official metrology laboratory	ISO expert groups, etc.	ISO 17043 Certified organization
Number of participants	Less than 10	At least 10	Up to several thousand
Number of samples	4 or 5	4 or 5	Several/year
Number of measures/samples	Often 5 or 6	At least 2	1
Measurement method(s)	Combine as many methods as possible	Only one	Each lab has its own

11.3 Education about Uncertainty

The practical problems that physicians, epidemiologists, researchers, analysts, engineers, and technicians solve every day have not a single solution but, in most cases, multiple solutions. The latter is documented by incomplete and uncertain data.

During high-school or university curriculum, to find the *unique* solution of an exercise, all useful elements are present in its statement. Unfortunately, this is not case in real life, and decisions must be made when only partial information is available. There is, therefore, a lack of training in measurement uncertainty in scientific curricula.

The classical (old) calculation of the maximum error, which used to be required of students after the sessions of practical work, can no longer be used anymore as a basis, considering the current evolution of knowledge in analytical science. Some initiatives exist to introduce this concept of MU at various levels of training, but it remains very fragmented. It can also be pointed out that the word “uncertainty” is inappropriate because it is confused with indecision, which is further emphasized by the famous precautionary principle.

However, we have tried to show how this parameter, inherent to any measurement, can reduce indecision. Will analysts soon report their results in the form, not as a single value but as a range of possible values, of a coverage interval? Throughout this book, we have presented this proposal as a necessary improvement for better interpretation of laboratory results and for decision-making. Paradoxically, end-users and prescribers may be reluctant and perceive this as inconvenient. It will therefore be necessary to explain and demonstrate the benefits to them and how they can integrate MU into their risk estimation.

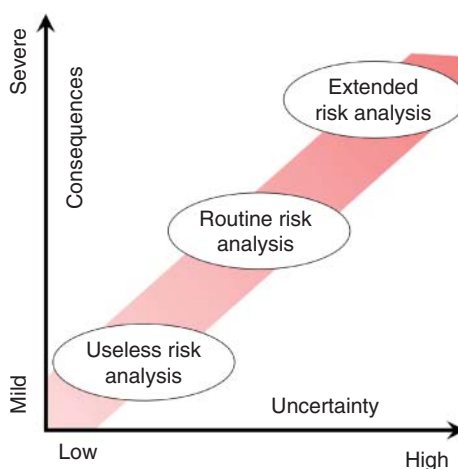
11.4 Risk Analysis

Several times, we have argued that MU knowledge makes it easier to make decisions because it gives an idea of the risk involved. The practicality of this proposal was based on providing a result in the form of a coverage interval, defined as “containing the set of true values of a measurand with a given probability, based on the available information.”

In all the examples presented, the coverage intervals included 95% of the possible true values of the measurand. In other words, this choice assumes that a percentage of possibly acceptable and *good* values are not considered as conforming. If the interval is used unilaterally, at most 2.5% of the values are rejected. Finally, the risk of a wrong decision would then be 2.5%. The question is to decide whether this risk is acceptable.

The choice of the 5% risk probability is very traditional. For example, a sampling plan is traditionally declared efficient when there is only a 5% chance of accepting a non-conforming lot. It is, therefore, a well-established industrial practice. But the use of analytical results is not limited to industrial batch release.

Figure 11.1 Relationship between risk consequences and measurement uncertainty. Source: Adapted from Yoe [3].



For example, for official food control, sanitary limits are established with large safety margins, and this probability could be increased to 10% (bilaterally) without significant risk. But when it comes to a trial or a suspicion of doping and when the life or the reputation of a suspect is at stake, can we accept the risk of being wrong five times out of 100?

The starting point for considering consequences is risk analysis. It applies to many areas, such as health safety, natural disasters, public health, or occupational accidents, all areas where analytical sciences are involved. For example in the case of food risk analysis, there are standardized or official documents outlining the procedure to be followed [2]. In other fields, clear answers to this question must also be the subject of a consensus specific to the analytical field.

A possible guide for this reflection can be drawn from Figure 11.1 adapted from a book on decision-making under uncertainty [3]. As this reference is primarily concerned with economic decision-making, this figure is concerned with uncertainty in general, not just measurement uncertainty.

According to this scheme, if we take the example of the release of a batch based on an analytical measurement, it is up to the decision-maker to know, according to the MU provided by the laboratory – undoubtedly including the sampling uncertainty – what type of risk analysis he must conduct. It is then a matter of identifying the hazards and their effects and quantifying them in terms of probability.

11.5 Harmonization of MU Estimation Procedures

The last question that only analysts can answer is the reduction in the number of approaches proposed in the literature for estimating MU in the analytical sciences. It seems obvious that this plethora can introduce unfair commercial practices and confuse users. Throughout the numerous examples presented, we have tended to always relate to the method accuracy profile (MAP), developed in the 2000s. This

approach is empirical, because the organization of data collection and the data processing were initially intended for the method validation and finally turns out to be extremely informative.

It allows us to answer a complete set of questions such as how to estimate the analytical part of MU, which together with the sampling part, is often the major part of the MU. It also makes it possible to predict the number of measurements needed and so be able to budget for validation. Furthermore, we have also explained how, historically, operation procedure standardization has led to a profound improvement in confidence in analytical results. It seems obvious then that a similar approach can also be applied to the calculation of MU.

Several documents, standards, guides, and directives already exist, as listed in the bibliography. They should not be confused with the standards produced by the metrology organizations alone, which often concern physical measurements. However, unlike analytical method standards, which are always accompanied by an experimental contribution in the form of an interlaboratory comparison, standards describing MU estimation methods are not accompanied by a practical, collaborative, and comparative application.

In conclusion, improving the interpretation of analytical results still requires a lot of scientific and technical work. This book aims to make progress along this path, but it certainly does not claim to have reached the end. Only collaborative work, in which analysts know how to adopt these new concepts, has a chance of success. Moreover, many papers proposing the use of MU in decision-making remain theoretical and do not show the practical consequences of this approach. We have therefore tried to provide examples of how MU can be used in practice.

References

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- 3 Yoe, C.E. (2019). *Principles of Risk Analysis: Decision Making under Uncertainty*. Boca Raton, FL: Taylor and Francis/CRC Press.